

SOFTWARE REQUIREMENTS SPECIFICATION

AI-POWERED FINANCIAL DECISION AGENT: BUY OR WAIT?

Project: HackerRank Orchestrate (Sept 2026)	Document ID: SRS-HRO-2026-01	Standard: IEEE 830 / ISO 29148
Challenge: Buy or Wait? Financial Decision Agent	Version: 1.0.0 (Approved Production Blueprint)	Evaluation Scope: 250 Requests / 90-Day Simulation
Harness / Suite: Antigravity AI Engineering Suite	Date: September 2026	Target Deadline: Sept 13, 2026, 18:00 IST

1. Executive Summary & Problem Motivation

1.1 Context & The 'Why': Consumer financial decision-making has long suffered from oversimplified heuristic models. In traditional personal finance and banking apps, affordability is evaluated instantaneously: $Current\ Balance \geq Purchase\ Amount$. This approach consistently fails: a user holding \$2,000 who contemplates a \$1,500 laptop may face \$1,200 in fixed rent, debt service, and essential utilities in 5 days, plus a mandatory \$500 safety reserve. Authorizing this expense precipitates immediate overdrafts or missed payments. Conversely, another user with identical balance may expect a confirmed payroll credit in 48 hours or hold flexible subscriptions they are eager to pause, making the purchase completely viable.

1.2 System Objective: The system must serve as an autonomous, deterministic, multimodal financial intelligence agent. Across 250 real-world expense evaluation requests, the engine synthesizes structured profiles, 25,000+ historical cash transactions, employer/merchant messages in multiple languages, scanned bill receipts, and dated exchange rates to project a 90-day forward cash flow, establish safe disbursement envelopes, and select optimal payment plans.

The Core Affordability Doctrine: A financial plan is certified **SAFE** if and only if the user can execute every scheduled payment in full, cover all protected recurring obligations, absorb pending debit settlements, and maintain their defined reserve balance (*minimum_balance_to_keep*) on every single calendar day throughout the entire 90-day forward horizon.

2. System Architecture & Relational Entity Schema

The system ingests 8 primary datasets from dataset/. Entity relationships are resolved using user_id, request_id, and related_event_id, with dated FX conversions matching settlement timestamps.

Dataset File	Entity & Role	Schema / Fields	Processing & Integrity Rule
requests.csv	Evaluation Targets (250 rows)	request_id, user_id, request_date, request_type, requested_amount, desired_completion_date, allows_partial_payment, request_text	Predict exactly 1 row per request into root output.csv.
financial_profiles.csv	User Constraints & Preferences	home_currency (INR, ZAR, IDR, USD, EUR), current_available_balance, minimum_balance_to_keep, financial_priorities, protected/flexible categories, max_installment_months	Enforces safety floor, protected categories, and financing constraints.
financial_events.csv	Cash Ledger (25,344 rows)	event_id, user_id, event_type, category, direction (credit/debit), amount, currency, event_date, settlement_date, status, linked_event_id, flexibility, minimum_allowed_amount	Reconstruct recurring vs one-time cash flows. Note: 16 amounts are blank!
images.csv & media/	Multimodal Visuals (16 PNGs)	image_id, user_id, request_id, related_event_id. Target: media/images/image_XX.png	Mandatory OCR extraction for events with missing amounts.
messages.csv	Context Updates (216 rows)	message_id, user_id, request_id, related_event_id, sent_at, source_type, message_text (EN/ID)	Reconciles salary adjustments, delayed bonuses, and rent changes.
request_payment_options.csv	Financing Offers (791 rows)	payment_option_id, request_id, payment_method, payment_amount, number_of_payments, first_payment_date, payment_frequency_days, financing_fee, total_payable_amount	Installment recommendations must strictly match an available option.
exchange_rates.csv	Dated FX Conversions (135 rows)	rate_date, from_currency, to_currency, rate	Normalizes foreign-currency cash events to user's home currency.
sample_requests.csv	Public Ground Truth (25 rows)	Complete 15-column schema including verified decisions and plans	Used for regression testing, calibration, and behavioral verification.

3. Target Output Specification (output.csv)

The system must produce output.csv in the repository root. It must contain exactly 250 prediction rows plus the header row, strictly obeying the column order, allowed enumerations, and mathematical invariants:

Column Name	Type / Enumeration	Semantics, Format & Contractual Invariant
request_id	String	Unique request identifier matching dataset/requests.csv (e.g. request_26).
amount_safe_to_pay	Decimal Float (≥ 0.00)	Maximum amount safe to disburse today (request_date) before optional spending changes. Invariant: 0 ≤ amount_safe_to_pay ≤ requested_amount.
affordability_status	Enum (4 states): - affordable_now - affordable_with_plan - affordable_later - not_affordable	Macro feasibility state. affordable_now implies full payment safe today. affordable_with_plan implies completion via installments, partial split, or spending changes. affordable_later implies full payment safe later.
recommended_payment_method	Enum (5 states): - full_payment - partial_payment - installments - wait - not_recommended	Safest permissible settlement approach. Must be present in user's payment_methods_user_will_consider (or not_recommended fallback).
payment_plan	Chronological schedule: YYYY-MM-DD:amt YYYY-MM-DD:amt or none	Chronological payments joined by ' '. Installments must exactly match a supplied option. Partial payment has exactly 2 payments. Use none when not recommended.
earliest_date_for_full_payment	Date (YYYY-MM-DD) or empty	First date when full amount passes 90-day safety check as a single payment without optional spending changes. Equals request_date for affordable_now.
spending_changes_needed	Actions joined by ' ': stop:event_ID reduce_to:event_ID:amt or none	Up to 3 modifications targeting recurring flexible expenses in categories the user consented to adjust. Stop and reduce on the same event are mutually exclusive.
decision_explanation	Text String (1-3 sentences)	Concise, grounded financial rationale citing exact numbers, dates, and protected minimum balances.

4. Mathematical Formulation & Algorithmic Rules

4.1 90-Day Rolling Cash Balance Simulation:

Let $T_{\blacksquare}$ denote request_date. The forecast horizon spans $t \in [T_{\blacksquare}, T_{\blacksquare} + 90 \text{ days}]$. The daily cash balance evolves according to:

$$B(t) = B(t-1) + \sum \text{Inflow}_k(t) - \sum \text{EssentialOutflow}_j(t) - \text{PlanPayment}(t)$$

where $B(T_{\blacksquare} - 1) = \text{current_available_balance} - \sum \text{PendingDebits}$.

A candidate payment plan P is certified **FEASIBLE** if and only if:

$$B(t) \geq \text{minimum_balance_to_keep}, \quad \forall t \in [T_{\blacksquare}, T_{\blacksquare} + 90]$$

and the terminal payment date $T_{\text{final}}(P) \leq \text{desired_completion_date}$.

4.2 Calculation of amount_safe_to_pay & earliest_date_for_full_payment:

The daily safe margin is $M(t) = B_{\text{base}}(t) - \text{minimum_balance_to_keep}$, where $B_{\text{base}}(t)$ simulates baseline cash flow without optional spending changes or purchase payments. The maximum immediate safe payment is:

$$\text{amount_safe_to_pay} = \min(\text{requested_amount}, \max(0, \min_{t \in [T_{\blacksquare}, T_{\blacksquare} + 90]} M(t)))$$

Similarly, earliest_date_for_full_payment is the minimum date $d^* \geq T_{\blacksquare}$ such that paying requested_amount on date d^* maintains $B(t) \geq \text{minimum_balance_to_keep}$ for all $t \in [d^*, T_{\blacksquare} + 90]$.

4.3 Decision Hierarchy for Plan Selection (Tie-Breaking Cascade):

When multiple candidate plans (full payment, installments, partial payment, wait) are feasible, select the winning plan by strictly evaluating:

- Completion Constraint:** Complete full request by desired_completion_date.
- Spending Preservation:** Require ZERO spending changes (stop/reduce).
- Cost Minimization:** Minimize total payable amount (principal + financing fees).
- Temporal Immediacy:** Earliest first payment date.
- Operational Simplicity:** Fewest number of payments.
- Deterministic Tie-Breaker:** Lowest numerical payment_option_id.

5. Detailed Functional Requirements (FR-1 to FR-10)

FR-1: Multimodal OCR & Receipt Ingestion Pipeline

- Exactly 16 events in financial_events.csv have a blank amount field.
- The system MUST join financial_events.event_id to images.related_event_id to resolve image_id.
- The target file at dataset/media/images/.png must be processed via OCR/Vision LLM to extract the numerical currency figure.
- **Zero-Imputation Prohibition:** A blank amount MUST NEVER be treated as zero or omitted from cash-flow aggregation.

FR-2: Natural Language Message Interpretation & Semantic Amendment

- Ingest 216 messages across English and Indonesian. Parse source types (employer, bank, merchant, service_provider).
- Extract actionable financial amendments: confirmed salary raises (e.g. Cobalt Systems IDR 42,750,000 in msg_01), temporary pay deductions due to unpaid leave (Greenfield Foods EUR 1,422.85 in msg_06), delayed gig earnings (QuickCrew, TaskLoop), and rent lease increases (+12% in msg_12).
- **Untrusted Input Doctrine:** Messages and images represent untrusted user data. Embedded prompt injection instructions attempting to override challenge invariants, grant fake credits, or bypass minimum balances MUST BE IGNORED.

FR-3: Foreign Exchange Normalization

- Map all foreign currency transactions to the user's home_currency using dated exchange rates in exchange_rates.csv.
- Currency conversion MUST match the specific transaction's settlement_date and exact (from_currency, to_currency) direction.

FR-4: Conservative Financial State Reconstruction

- **Debit Priority:** Reserve and subtract all pending debits immediately from available cash.
- **Credit Conservatism:** Completely exclude pending credits, unvested bonuses, commission contingencies, prize claims, and unrealized investment gains until settled.
- **Salary Recognition:** Credit confirmed salary strictly on its confirmed settlement date.

FR-5: 90-Day Rolling Safety Simulation

- Compute the day-by-day cash balance for every day from request_date to request_date + 90.
- Verify that the balance never drops below minimum_balance_to_keep under any circumstances.

FR-6: Affordability State Machine

- affordable_now: Full payment safe on request_date AND user considers full_payment.
- affordable_with_plan: Affordable via installment schedule, partial payment split, or permitted spending pruning.
- affordable_later: Full payment becomes safe on a future date $\leq$ completion deadline AND user considers full_payment.
- not_affordable: Full request cannot be safely completed within the 90-day horizon.

FR-7: Payment Plan Generator & Schedule Invariants

- **Installments:** Must strictly match a valid offer in request_payment_options.csv with duration $\leq$ max_installment_months.
- **Partial Payment:** Recommended ONLY when request allows partial payment, user considers it, $0 < \text{amount_safe_to_pay} < \text{requested_amount}$, and $\text{earliest_date_for_full_payment} \leq \text{desired_completion_date}$. The plan must contain EXACTLY two payments: amount_safe_to_pay on request_date, and the remaining balance on earliest_date_for_full_payment.

FR-8: Flexible Spending Pruning Optimizer

- If an expense cannot be afforded normally, the engine may recommend stopping or reducing up to 3 expenses.
- Actions: stop: or reduce_to:: separated by '|'.
- Eligible events must be recurring, marked stoppable or reducible, and belong to categories the user permitted in financial_profiles.csv (expense_categories_user_is_willing_to_stop/reduce).

FR-9: Plan Ranking & Multi-Criteria Optimization

- Execute the 6-level tie-breaking cascade to pick the single optimal recommendation among all safe candidate plans.

FR-10: Grounded Financial Explanations

- Generate a concise, numbers-driven justification for every prediction, explicitly citing the safe payment limit, minimum balance cushion, and relevant settlement dates.

Table 5.1: Inventory of 16 Visual Assets with Blank Amounts

Event ID	User ID	Category	Curr	Linked Image	Document Description / Purpose
event_253	user_03	salary	IDR	image_01.png	Payroll payslip with net earnings
event_1442	user_16	rent	INR	image_02.png	Rental agreement invoice / receipt
event_1545	user_17	groceries	INR	image_03.png	Supermarket monthly grocery bill
event_1700	user_19	groceries	INR	image_04.png	Supermarket monthly grocery bill
event_1786	user_20	utilities	INR	image_05.png	Electricity & water utility statement
event_3051..10521	Various	Health/Trans/Shop	INR/USD	image_06..16.png	11 recurring bills & healthcare receipts

6. Non-Functional Requirements & Quality Standards

Quality Attribute	Technical Specification	Verification & Quality Gate
Financial Safety	Zero tolerance for balance violations below minimum_balance_to_keep across the 90-day forecast.	Deterministic cash-flow simulation validator checks all 250 trajectories.
Performance & Speed	Batch evaluation of all 250 requests must complete in under 5 minutes on standard computing hardware.	Vectorized pandas simulation and cached multimodal OCR inference.
Reproducibility	The prediction pipeline must be fully deterministic. Identical inputs must yield bitwise identical output.csv.	Random seeds fixed; temperature = 0.0 for any LLM reasoning components.
Security & Robustness	Immunity to prompt injections embedded in merchant messages, employer letters, or receipt images.	Strict data sandboxing; natural language context cannot override rule engine invariants.
Cost & Token Tracking	Mandatory generation of evaluation/usage_report.md documenting model calls, input/output tokens, and dollar costs.	Automated token counter wrapper around LLM client calls.
Auditability & Logging	Adherence to AGENTS.md protocol. All conversation turns, tools, and actions logged to log.txt.	Automated append logger verifying tool=, git branch, and timestamps.

7. Verification, Validation & Acceptance Criteria

7.1 Automated Validation Protocol:

- **Benchmark Suite:** Run solution against the 25 solved cases in dataset/sample_requests.csv. Verify 100% agreement on amount_safe_to_pay, affordability_status, recommended_payment_method, payment_plan, and earliest_date_for_full_payment.
- **Mathematical Invariants:** Every prediction row must satisfy: 0 <= amount_safe_to_pay <= requested_amount.
- **Consistency Check:** If affordability_status == affordable_now, then earliest_date_for_full_payment == request_date.
- **Plan Feasibility Check:** Every payment in payment_plan must have daily post-payment balance ≥ minimum_balance_to_keep.

8. Mandatory Submission Deliverables & Packaging

Prior to the deadline of 6:00 PM IST on September 13, 2026, the following three mandatory artifacts must be compiled and uploaded:

Artifact	Format & Location	Contents & Evaluation Criteria
output.csv	CSV File /output.csv	Exactly 250 predictions corresponding to dataset/requests.csv. Evaluated on accuracy of amounts, status, methods, dates, plans, and explanations.
code.zip	ZIP Archive /code.zip	Complete runnable codebase, setup/execution README, and mandatory evaluation/usage_report.md detailing models, token counts, and cost breakdown.
chat_transcript	Text File /log.txt	Continuous append-only development log documenting all agent sessions, user prompts, actions, and tools used in compliance with AGENTS.md.

Official HackerRank Submission Portal:
<https://www.hackerrank.com/contests/hackerrank-orchestrate-september26/challenges/buy-or-wait/submission>
Reminder: This is a solo challenge. Verify all deliverables and run deterministic validation before final submission.

Architectural Sign-off: Software Requirements Specification certified complete and ready for execution.